HDOC-53

Product of Array Elements Except Self Using Pointers

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array nums[]. Pass the array to product_except_self() using its address and use pointers to pass the arrays. Create answer[] of the same size and store at answer[i] the product of all elements except nums[i], without using division. Display answer[] in original order separated by commas. The prefix and suffix products are guaranteed to fit in a 32-bit integer.

Algorithm

1. Read n and the n integer elements into nums[].
2. Create answer[n]. Pass nums and answer to product_except_self() as pointers.
3. Set prefix = 1. Traverse left to right: store prefix in answer[i], then multiply prefix by nums[i].
4. Set suffix = 1. Traverse right to left: multiply answer[i] by suffix, then multiply suffix by nums[i].
5. Display answer[] in the original order, separated by commas.

C Program

```
#include <stdio.h>

void product_except_self(int *nums, int n, int *answer)
{
    int prefix = 1, suffix = 1;
    int i;

    for (i = 0; i < n; i++) {
        *(answer + i) = prefix;
        prefix = prefix * (*(nums + i));
    }

    for (i = n - 1; i >= 0; i--) {
        *(answer + i) = (*(answer + i)) * suffix;
        suffix = suffix * (*(nums + i));
    }
}

int main(void)
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    int nums[n], answer[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", nums + i);

    product_except_self(nums, n, answer);
```

```
    printf("Answer: ");
    for (i = 0; i < n; i++) {
        printf("%d", *(answer + i));
        if (i < n - 1)
            printf(", ");
    }
    printf("\n");

    return 0;
}
```

Sample Output

```
Enter the size of the array: 4
Enter 4 elements: 1 2 3 4
Answer: 24, 12, 8, 6
```

Test Cases

Test Case	Input Array	Expected Output	Purpose
1	[1, 2, 3, 4]	[24, 12, 8, 6]	Normal positive values
2	[2, 3, 4]	[12, 8, 6]	Three elements
3	[1, 0, 3, 4]	[0, 12, 0, 0]	One zero
4	[-1, 2, -3, 4]	[-24, 12, -8, 6]	Negative values
5	[5]	[1]	Single element

Result

The program successfully calculates the product of all array elements except the current element using prefix and suffix products, pointers, and no division.